

Stochastic

Probability, sampling & Monte Carlo. All in SQL.



Install & load

```
INSTALL stochastic FROM community;
LOAD stochastic;
```

01 One naming pattern

```
dist_<family>_<operation>(params..., x)
```

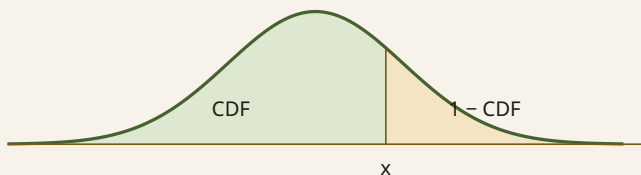
Distribution parameters come first. Add x for density or cumulative probability, p for a quantile, and neither for sampling or properties.

Operation suffix	Returns
sample	One random draw
pdf	Density / discrete mass ¹
cdf	Left tail: $P(X \leq x)$
cdf_complement	Right tail: $P(X > x)$
quantile	Inverse CDF at p
quantile_complement	Inverse right tail at p
log_pdf	Natural log of PDF
log_cdf	Natural log of CDF
log_cdf_complement	Natural log of right tail
hazard	PDF / right tail
chf	$-\ln(\text{right tail})$

Properties: no trailing x or p

mean, median, mode, variance, stddev, skewness, kurtosis, kurtosis_excess, range, support. Availability depends on the family; range and support return two-element arrays.

02 Read the probability



The CDF is the shaded area left of x. Its complement is the area to the right. For continuous distributions, $\text{pdf}(x)$ is a density, not $P(X = x)$.

Tail arithmetic

For discrete counts: $P(X \geq k) = \text{cdf_complement}(\text{params} \dots, k-1)$. Call the complement directly rather than subtracting a CDF from 1.

03 Normal: quantiles & CDFs

Parameters are mean and standard deviation.

```
SELECT dist_normal_quantile(0, 1, 0.95);
-- 1.6448536269514729
SELECT dist_normal_cdf(0, 1, 1.96);
-- approximately 0.975
```

04 Count probabilities

```
-- P(X ≥ 5), Poisson(rate = 2.5)
SELECT dist_poisson_cdf_complement(2.5, 4);
-- 0.10882198108584878

-- P(X ≤ 5), Binomial(n = 10, p = 0.3)
SELECT dist_binomial_cdf(10, 0.3, 5);
-- 0.9526510126000001
```

05 Generate → evaluate → average

Estimate π from uniform points in the square. Materialize draws so every expression reuses the same sampled coordinates.

```
WITH draws AS MATERIALIZED (
  SELECT
    dist_uniform_real_sample(-1, 1) AS x,
    dist_uniform_real_sample(-1, 1) AS y
  FROM range(100000)
)
SELECT 4.0 * AVG(
  (x*x + y*y ≤ 1) :: INTEGER
) AS pi_estimate
FROM draws;
```

Result varies by run and approaches π as the sample size grows.

Keep draws for reproducibility

`_sample` has no seed argument. Save draws with `CREATE TABLE ... AS SELECT ...` and reuse that table for repeatable analysis.

¹ `uniform_int` uses inclusive integer endpoints; see page 2. Discrete quantiles use family-specific rounding.

[Full docs & examples ↗](#) [Source ↗](#)

Stochastic

The distribution directory



Use the family token in `dist_<family>_<operation>`. Parameters below are listed in call order; append `x` or `p` when the operation needs it.

06 Continuous families

Family token	Parameters / interpretation
beta	alpha, beta Positive shapes; proportions
cauchy	location, scale Heavy tails; no mean / variance
chi_squared	df Positive degrees of freedom
exponential	rate Mean waiting time = 1/rate
extreme_value	location, scale Gumbel; block maxima
fisher_f	df1, df2 Ratio of scaled variances
gamma	shape, scale Both positive; mean = shape × scale
laplace	location, scale Symmetric, sharper peak
logistic	location, scale Symmetric, heavier tails
lognormal	meanlog, sdlog Parameters of $\ln(X)$
normal	mean, stddev Location and spread
pareto	scale, shape * Lower bound and tail shape
rayleigh	scale Nonnegative magnitudes
students_t	df Standard t; positive df
uniform_real	min, max min < max
weibull	shape, scale Both positive; failure times

* Pareto's runtime order is (scale, shape). Fixed v1.5 labels match this order. Older 234c881 labels incorrectly say (shape, minimum).

07 Discrete sampling families

Family token	Parameters / interpretation
bernoulli	p A Boolean draw; $0 \leq p \leq 1$
binomial	trials, p Successes in a fixed trial count
geometric	p Failures before first success
negative_binomial	successes, p Failures before target successes
poisson	rate Positive expected event count
uniform_int	min, max Integer draws; both ends included

Integer uniform: inclusive endpoints

Fixed v1.5 analytics match integer sampling. For bounds 1 to 6, PMF = 1/6, CDF at 3 = 0.5, mean = 3.5 and variance = 35/12. Older 234c881 uses incorrect continuous formulas and truncates moments.

08 Make a synthetic dataset

```
CREATE TABLE synthetic_users AS
SELECT
  i AS user_id,
  dist_normal_sample(170, 8)
    AS height_cm,
  dist_poisson_sample(3.5) AS sessions,
  dist_bernoulli_sample(0.12)
    AS converted
FROM range(10000) AS t(i);
```

Each row receives fresh draws. Bernoulli returns BOOLEAN; other discrete samplers return BIGINT; continuous samplers return DOUBLE.

09 Check the installed surface

```
SELECT function_name, parameters
FROM duckdb_functions()
WHERE function_name LIKE 'dist_normal_%'
ORDER BY function_name;
```

Use positive scales and standard deviations. Properties may be undefined at some parameter values; invalid inputs can raise SQL errors.

[Full docs & examples ↗](#) [Source ↗](#)